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E-420 VOLTAGE TRANSFORMERS

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of 500 kV voltage transformers, capacitor voltage transformer (CVT) in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards:

2.1 Indonesian National Standards

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60044 : Instrument transformers
- IEC 60060 : High-voltage test techniques
- IEC 60186 : Voltage transformers
- IEC 60270 : High-voltage test techniques – Partial discharge measurements
- IEC 60358 : Coupling capacitors and capacitor dividers
- IEC 60947 : Low-voltage switchgear and controlgear
- IEC 62155 : Hollow pressurized and un-pressurized ceramic and glass insulators for use in electrical equipment with rated voltages greater than 1000 V

2.3 National Electrical Manufacturers Association (NEMA)

- NEMA 107 : Method of Measurements of Radio Influence Voltage (RIV) of High Voltage Apparatus

3. OPERATING AND DESIGN CONDITIONS

The substation 500 kV voltage transformers shall be designed for outdoor operation as part of the 500 kV open air substation located in a tropical climate along

the seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for meteorological, seismic and environmental data of the site.

4. **DETAILED REQUIREMENTS**

4.1 **General**

The specified accuracy and burden ratings are the minimum requirements. The Contractor shall verify the ratings based on the applications and his proposed protective relays and instrument and propose the appropriate ratings.

Unless mentioned otherwise, the term capacitor voltage transformer (CVT) refers to the electromagnetic type voltage transformers with capacitance voltage divider, used in the 500 kV switchyard.

Strip heaters for condensation control shall be furnished in non-sealed housings in accordance with Technical Specification E-001.

Two (2) grounding terminal pads on diagonally opposite location shall be furnished for connections to ground grid with an ampacity not less than that of 240 mm² copper cable. Standard pads shall be furnished for connection to ground grid.

Secondary fuses or Miniature Circuit Breakers (MCBs) shall be provided with each CVT and shall be located such that they can be removed and replaced or reset while the primary is energized. The fuses or MCBs shall be located on or adjacent to each CVT with labels identifying function and phase color. Where a CVT is used for protective relays which respond to a blown fuse, a protective switch or similar device shall be provided which shall prevent the spurious relay operation due to such blown fuse and simultaneously provide an alarm.

Each CVT secondary shall be provided with a separate grounding link located at the transformer. Each circuit secondary shall be grounded at one (1) location only, preferably at the first panel entry at the control building.

Each CVT shall be equipped with a ferro-resonance damping circuit of proven design. The circuit shall be effective under all system conditions.

4.2 **Ratings**

(1) Nominal voltage level 500 kV

- (2) Maximum voltage level.....550 kV
- (3) Frequency 50 Hz
- (4) Secondary ratings:
 - a. Overall voltage ratio:500,000/ $\sqrt{3}$: 110/ $\sqrt{3}$ volts
 - b. Minimum number of secondary windings ...2
 - c. Minimum burden ratings each winding.....150 VA
 - d. Maximum ratio error0.25 %
 - e. Accuracy Class:
 - e.1 Protection.....3P
 - e.2 Instrumentation and Metering0.5
- (5) Phase-to-ground insulation:
 - a. Impulse withstand voltage (1.2 x 50 μ s) crest 1550 kV
 - b. 50 Hz (and wet*) withstand test voltage680 (-) kV
 - c. Switching impulse withstand test voltage 1175 kV
- (6) Radio Interference Voltage (RIV)

The maximum RIV at 1 MHz shall be:

 - a. Applied phase to ground voltage333 kV
 - b. RIV2,500 μ volts

Measurements shall be in accordance with NEMA 107 procedure. In marking the RIV test, radio noise readings shall be taken at various points to determine the break point.
- (7) Partial Discharge Test
 - a. Applied phase to ground voltage550 kV
 - b. Maximum discharge10 pico coulombs
5 μ volts

4.3 **Corona**

Surfaces in contact with live potential shall contain no projecting points or irregularities and they shall be designed for a minimum of visible corona or brush discharge when the transformer is completely assembled.

No visual corona shall exist at the applied voltage. Lights shall be turned off in the test bay, and visual start and stop corona voltages shall be recorded with the corresponding radio noise voltages on both ascending and descending voltage values. Photographs shall be taken for each recorded voltage value. NEMA 107 test procedures shall be used.

4.4 Porcelain

The Contractor shall meet the requirements of IEC 62155 for all porcelain used in these transformers and capacitors.

Porcelain insulator for CVTs shall be of brown color and shall be permanently marked for traceability by means that will not indent or affect porcelain surface.

4.5 Insulating Oil

Oil (if required) shall meet the requirements of Technical Specification E-001.

4.6 Nameplates

Nameplates shall be as per IEC 60186 and shall be provided clearly identifying individual winding classification for multiple secondary. Capacitor units shall be separately identified as per IEC 60358.

5. PERFORMANCE TESTING AND GUARANTEES

Voltage transformer shall be tested in accordance with IEC 60044 and 60186 and coupling capacitors shall be tested in accordance with IEC 60947.

The following additional tests shall be provided as type tests for each type of transformer or capacitor: If type on identical transformer has been done, then the certificate shall be submitted, and no repeated test is required.

- (1) Radio influence voltage tests on a complete transformer assembled in service as per this Specification and NEMA 107 along with a corona test.
- (2) Partial discharge test on a complete transformer of each type as per IEC 60270.
- (3) The Contractor shall verify that each type of voltage transformer shall meet the accuracy requirement of IEC 60044 and 60186 over the temperature range in Part 4, Section 2.16.
- (4) The Contractor shall measure at least one (1) pair of each type CVT being supplied for high frequency insert loss over the standard frequency range,

providing an insertion loss of the complete arrangement including matching transformers.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control CVTs shall conform to the requirements included in Part 4, Section 2, Subsection 2.7.

The quality assurance/quality control procedures specified shall include, but not be limited to, the following:

- (1) All routine tests specified in Clause 5 of this Specification.
- (2) All special type tests performed for the Specification as per Clause 5.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the following:

- (1) Shop Test Reports.
- (2) All other documentation required by applicable codes and standards.

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